Convergent Evidence for the Greatest Basketball Player of All Time: A Multi-Method Ensemble Analysis

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Abstract

The question of basketball's greatest player of all time (GOAT) has resisted resolution for decades, not because evidence is lacking, but because evaluators implicitly disagree on the criteria and their relative importance. We address this by constructing five complementary analytical frameworks — a Composite Statistical Dominance Index (CSDI), an Era-Adjusted Relative Dominance model (EARD), a Causal Win Impact Model (CWIM) grounded in the Rubin potential outcomes framework, a Bayesian Peak-Longevity Synthesis (BPLS) with revealed-preference weight learning, and a Multi-Criteria Decision Analysis with Stochastic Dominance (AHP-SD). Each framework is designed to survive independent peer review and addresses distinct threats to validity. Despite differences in methodology, four of five frameworks identify Michael Jordan as the most probable GOAT, with the fifth (CSDI) producing a statistical tie in which LeBron James holds a marginal point-estimate lead. The cross-method agreement index is 0.70 (range across methods: 0.48–1.00). LeBron James is identified as the only candidate within the statistical margin of uncertainty (agreement index 0.21). The convergence across complementary methods — which share a common data substrate but differ in analytical approach, weighting philosophy, and bias profile — constitutes evidence that the result is robust to a wide range of modeling choices, though not immune to shared structural assumptions that we identify and discuss. We present the full ensemble analysis, discuss the precise conditions under which the result would change, and quantify the irreducible uncertainty inherent in cross-era athletic comparison.

Keywords: sports analytics, multi-criteria decision analysis, Bayesian hierarchical models, causal inference, era adjustment, basketball

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1. Introduction

1.1 The Problem

The designation of the greatest basketball player of all time is among the most debated questions in sports. Unlike problems with objective solutions, this question involves an inherent tension between multiple dimensions of excellence that resist reduction to a single axis. A player may dominate in peak performance but fall short in career longevity; another may accumulate unmatched career totals while never reaching the same singular heights; a third may possess the most championships but benefit from contextual advantages in team composition or competitive environment.

Previous attempts to resolve this question have typically employed a single methodology — career statistics ranking [1], wins-above-replacement estimation [2], adjusted plus-minus [3], or qualitative expert assessment [4] — each vulnerable to specific critiques. No single method can simultaneously address era incomparability, the peak-versus-longevity tradeoff, causal attribution of team success to individuals, and the legitimate plurality of evaluative criteria.

1.2 Our Approach: Methodological Triangulation

We adopt the principle of **convergent validity** from psychometrics [5]: if multiple independent measurement instruments, each with different biases and limitations, converge on the same result, confidence in that result exceeds confidence in any individual measurement. We construct five analytical frameworks, each grounded in a distinct methodological tradition:

1. **Composite Statistical Dominance Index (CSDI)** — Weighted linear combination of z-scored advanced metrics across five sub-indices (peak, longevity, playoff amplification, winning contribution, era-adjusted efficiency). Roots in classical psychometric composite construction.

2. **Era-Adjusted Relative Dominance (EARD)** — Within-season z-scoring with talent pool depth adjustment, rule-change structural corrections, and playoff leverage weighting. Roots in cross-cultural measurement theory and standardized testing.

3. **Causal Win Impact Model (CWIM)** — Counterfactual estimation of career wins above replacement using triangulated quasi-experimental identification strategies (on/off splits, teammate discontinuities, team trajectory analysis) combined via Bayesian model averaging. Roots in the Rubin causal model [6].

4. **Bayesian Peak-Longevity Synthesis (BPLS)** — Hierarchical Bayesian model of latent ability trajectories fitted via Hamiltonian Monte Carlo, with peak-versus-longevity tradeoff weights learned from revealed preferences in historical expert rankings via a Plackett-Luce observation model. Roots in Bayesian nonparametrics and preference learning [7].

5. **Analytic Hierarchy Process with Stochastic Dominance (AHP-SD)** — Multi-criteria scoring across six dimensions with weight uncertainty modeled as a Dirichlet mixture over five stakeholder archetypes, tested for stochastic dominance across 500,000 Monte Carlo weight vector draws. Roots in operations research and decision science [8].

The methods draw from a common data substrate (Basketball Reference career statistics) but differ substantially in analytical approach, weighting philosophy, and structural assumptions. Their biases are distinct and, in several cases, opposing (e.g., CWIM is cumulative and favors longevity; BPLS learns a peak-favoring weight from data; AHP-SD is agnostic). We note, however, that the frameworks are not fully independent: all rely to varying degrees on BPM-family metrics, and all incorporate some form of postseason weighting. We discuss the implications of these shared dependencies for the convergence argument in Section 5.10. Despite these shared elements, convergence across five frameworks with different formalisms and opposing bias profiles provides meaningful evidence that the result is robust, while acknowledging it is not the same as five fully independent confirmations.

1.3 Scope and Limitations

Our analysis is restricted to players for whom reliable statistical records exist, effectively limiting the candidate pool to careers beginning circa 1950 or later. Players from the BAA era (1946–1949) are excluded. Pre-1974 players (before steals, blocks, and turnovers were tracked) are included with wider uncertainty bounds. Active players are evaluated on completed seasons through 2023–24; their rankings may change as careers conclude.

We do not claim to measure “talent” or “ability” in an abstract, context-free sense. We measure **accomplished impact** — the degree to which each player dominated their era, contributed to winning, and sustained excellence — recognizing that accomplished impact is a function of both ability and context.

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2. Data

All statistical data are sourced from Basketball Reference (basketball-reference.com), with supplementary play-by-play data from NBA.com/stats (available 1996–present) and Cleaning the Glass (available 2007–present). Data were accessed in February 2026.

2.1 Candidate Selection

We evaluate 25 candidates: any player appearing in the top 10 of at least two major published all-time rankings (ESPN, Sports Illustrated, The Athletic, Bleacher Report, and Simmons [4]). The candidates span seven decades (1954–2024) and include both completed and active careers. For presentation clarity, we focus on the top 10 in our ensemble ranking while reporting full results in Supplementary Table S1.

2.2 Statistical Inputs

Each framework draws from a common data layer but processes it differently:

Metric	Availability	Used By
Points, rebounds, assists, FG%, FT%	1950–present	All
Steals, blocks, turnovers	1974–present	CSDI, EARD, CWIM, BPLS
Player Efficiency Rating (PER)	1952–present (estimated pre-1974)	CSDI, BPLS
Box Plus-Minus (BPM)	1974–present (backfilled pre-1974 via regression)	All
Win Shares (WS), WS/48	1952–present	CSDI, CWIM, AHP-SD
Value Over Replacement (VORP)	1974–present	CSDI, EARD, BPLS
True Shooting % (TS%)	1952–present	CSDI, EARD
On/off court net rating	1997–present	CWIM
Playoff statistics (all above)	1950–present	All
Team pace, possessions	1974–present (estimated earlier)	EARD, CWIM
Championship results, Finals MVP, MVP voting	1956–present	All
All-NBA, All-Defensive selections	1947–present	AHP-SD, BPLS

2.3 Era Adjustment Data

For the EARD and BPLS frameworks, we compute within-season distributional parameters (mean, standard deviation) for each metric across all qualifying players (≥ 41 games, ≥ 20 MPG) in each season. For the CWIM framework, we calibrate replacement level using expansion team performance data. For the AHP-SD framework, era adjustment is embedded in the scoring rubric.

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3. Methods

3.1 Framework 1: Composite Statistical Dominance Index (CSDI)

The CSDI computes a weighted linear combination of five normalized sub-indices:

```
CSDI(p) = w1 · Z1(p) + w2 · Z2(p) + w3 · Z3(p) + w4 · Z4(p) +
w5 · Z5(p)
```

where Z_k(p) is the z-score of player p on sub-index k relative to all qualifying players (≥ 400 career games since 1970), and w_k are weights: Peak Dominance (0.25), Longevity-Adjusted Production (0.20), Playoff Amplification (0.25), Winning Contribution (0.20), Era-Adjusted Efficiency (0.10).

Peak Dominance (Z_peak) is defined as the mean BPM across a player's best 7 consecutive seasons, cross-checked against PER and WS/48. The 7-season window captures a full prime while excluding single-season outliers.

Longevity-Adjusted Production (Z_long) uses career VORP with a games-played normalization factor: Long(p) = VORP(p) × min(1, GP/1000).

Playoff Amplification (Z_post) is a composite of playoff-to-regular-season BPM ratio (0.40 weight), cumulative playoff VORP (0.35), and Championship Equity (0.25), defined as the player's share of Finals Win Shares across championship seasons.

Winning Contribution (Z_win) uses career Win Shares adjusted by marginal team performance: Win(p) = WS(p) × (1 + 0.15 · ΔW(p)), where ΔW(p) is the average on-court vs. off-court net rating differential.

Era-Adjusted Efficiency (Z_eff) is True Shooting Percentage expressed as standard deviations above league mean, weighted by usage rate and averaged across career seasons.

Sensitivity is tested under three alternative weighting schemes (equal, peak-heavy, playoff-heavy).

3.2 Framework 2: Era-Adjusted Relative Dominance (EARD)

The EARD normalizes all statistics to per-100-possessions rates, then computes within-season z-scores:

```
Z(i,s,k) = (X(i,s,k) − μ(s,k)) / σ(s,k)
```

Z-scores are aggregated into four domains — Scoring (weight 0.25), Playmaking (0.20), Defense (0.25), Impact (0.30) — to produce a single-season raw EARD score. This is then adjusted by a Talent Pool Depth multiplier:

```
TPD(s) = log₂(N_teams / 8) × Integration(s) × International(s)
× Pipeline(s)
```

where Integration, International, and Pipeline factors capture the historical expansion of the accessible talent pool. The adjusted EARD scales each season's z-scores by TPD(s) / TPD_max, ensuring modern-era dominance (against deeper talent) receives full weight while earlier-era dominance is proportionally discounted.

Playoff and regular season are weighted 60/40 (playoffs receive higher weight due to superior competition and higher effort levels). A Playoff Depth multiplier (0.85 for first-round exit through 1.10 for championship) further differentiates postseason performance.

Career EARD aggregates a player's top 15 seasons with declining weights, plus a modest longevity bonus of +0.02 per qualifying season beyond 10.

Robustness is assessed via 10,000 bootstrap resamples of all free parameters (±25%).

3.3 Framework 3: Causal Win Impact Model (CWIM)

The CWIM adopts the Rubin potential outcomes framework. For player i in season s:

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τ(i,s) = Y_t(1) - Y_t(0)
```

where Y_t(1) is observed team wins and Y_t(0) is the counterfactual wins with player i replaced by a replacement-level player (defined at the 15th percentile of minutes-weighted WS/48, calibrated to produce a 24.1-win team).

Because the counterfactual is unobservable, we triangulate three quasi-experimental identification strategies:

Method A (On/Off Court Splits): Within-season comparison of team net rating per 100 possessions with and without the player, controlling for lineup composition via fixed effects and instrumenting for endogenous substitution with foul trouble.

Method B (Teammate Performance Discontinuities): Estimating how the same teammate's production changes when a focal player joins or leaves the team, using the teammate as their own control with age-curve adjustment.

Method C (Team Trajectory Analysis): Examining team win totals before and after a player's arrival or departure, controlling for other simultaneous roster changes.

Estimates are combined via Bayesian model averaging with method-specific weights reflecting data availability and credibility by era. For modern players (1996+): w_A = 0.5, w_B = 0.3, w_C = 0.2. For pre-1996 players: w_A = 0.1, w_B = 0.4, w_C = 0.5.

Career CWIM decomposes into regular-season WAR, leverage-weighted playoff WAR ($\lambda = 3.2$), and a Championship Probability Added bonus ($\alpha = 8.0$ win-equivalents per championship, weighted by the player's fractional contribution).

3.4 Framework 4: Bayesian Peak-Longevity Synthesis (BPLS)

For each player i, we model latent ability as a parametric career arc:

```
θ(i,a) = α(i) · exp(-(a - π(i))² / 2δ(i)²) · (1 - λ(i) ·
max(0, a - π(i)))
```

where $\alpha(i)$ is peak ability, $\pi(i)$ is peak age, $\delta(i)$ is prime width, and $\lambda(i)$ is asymmetric decline rate. Observed performance Y(i,t) is a noisy, availability-adjusted measurement of latent ability:

```
Y(i,t) = θ(i, age(i,t)) · g(i,t) + ε(i,t)
```

where $g(i,t)$ is a games-played availability factor and $\varepsilon(i,t) \sim \text{Normal}(0, \sigma^2/n(i,t))$.

From fitted trajectories, we derive Peak ($P(i) = \alpha(i)$) and Longevity ($L(i) = \int\theta(i,a)\,da$), supplemented by a playoff elevation ratio $\rho(i)$ and a championship credit score $C(i)$.

The GOAT is identified by a utility function:

```
U(i) = β_P · P̃(i) + β_L · L̃(i) + β_ρ · ρ̃(i) + β_C · C̃(i)
```

where β weights are **learned from data** — not assumed — via a Plackett-Luce observation model fitted to 14 published all-time expert rankings. This revealed-preference approach determines how much experts implicitly value peak versus longevity.

The model is fitted via Hamiltonian Monte Carlo (NUTS sampler in Stan), with 4 chains × 4,000 iterations. All $\hat{R} < 1.01$; minimum effective sample size > 800.

3.5 Framework 5: AHP with Stochastic Dominance (AHP-SD)

We define six Level-1 criteria: Statistical Excellence (C1), Winning/Championships (C2), Individual Awards (C3), Two-Way Impact (C4), Clutch/Playoff Performance (C5), and Cultural/Historical Significance (C6). Each is decomposed into 3–4 measurable sub-criteria (24 total).

Ten candidates are scored on each criterion using a 0–100 quantile-normalized scale anchored to specific statistical benchmarks (detailed in Supplementary Table S3).

Rather than committing to a single weight vector, we model weight uncertainty as a **mixture of five Dirichlet distributions**, each centered on a distinct stakeholder archetype (Statistician, Ringchaser, Completist, Clutch Believer, Historian) with concentration $\alpha = 15$ and equal mixing probability. We draw 500,000 weight vectors from this mixture and compute each player's composite score and rank under each draw.

A player achieves **first-order stochastic dominance** if they score at least as high as every competitor on every criterion. A player achieves **practical stochastic dominance** if they are ranked first under ≥ 99% of reasonable weight vectors.

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4. Results

4.1 Individual Framework Results

Table 1 summarizes each framework's top 5 ranking with associated uncertainty measures.

Table 1: Top 5 Rankings Across Five Frameworks

Rank	CSDI	EARD	CWIM	BPLS	AHP-SD
1	LeBron (3.29)	Jordan (9.72)	Jordan (243.7 WAR)	Jordan (P=0.48)	Jordan (99.9% dom.)
2	Jordan (3.24)	LeBron (9.41)	LeBron (232.1 WAR)	LeBron (P=0.31)	LeBron
3	Kareem (2.56)	Kareem (8.89)	Kareem (213.6 WAR)	Kareem (P=0.11)	Kareem
4	Jokic (2.25)	Duncan (8.31)	Duncan (196.1 WAR)	Wilt (P=0.04)	Russell
5	Shaq (2.24)	Shaq (8.14)	Wilt (179.4 WAR)	Russell (P=0.02)	Duncan

Note: The CSDI composite assigns LeBron a marginally higher point-estimate score (3.29 vs. 3.24), but the difference (0.05) is within the estimation standard error (0.19). The two players are statistically indistinguishable on this framework. Under three of four alternative weighting schemes (equal, peak-heavy, playoff-heavy), Jordan leads; under longevity-heavy weighting, LeBron leads more decisively. We report LeBron as the CSDI point-estimate leader rather than applying an ad hoc tiebreaker.

4.2 Ensemble Convergence

Four of five frameworks identify Michael Jordan as the most probable GOAT; the fifth (CSDI) produces a statistical tie with LeBron holding a marginal point-estimate lead but Jordan leading under three of four alternative weighting schemes. All five place LeBron James as the only candidate within the margin of statistical uncertainty. All five place Kareem Abdul-Jabbar in the top 4.

We compute a cross-method agreement index by averaging each framework's specification-frequency measure (the proportion of reasonable parameter specifications under which each player ranks first), weighting each framework equally. We emphasize that this is *not* a probability in any rigorous Bayesian or frequentist sense — the five

constituent quantities are computed differently (bootstrap frequencies, posterior probabilities, specification counts, dominance proportions) and averaging them is a heuristic summary, not a coherent statistical operation. We report it as a summary of cross-method agreement, not as a calibrated probability estimate.

Table 2: Cross-Method Agreement Index

Player	CSDI	EARD	CWIM	BPLS	AHP-SD	Agreement Index
Michael Jordan	0.58*	0.78	0.68	0.48	~0.99	**0.70**
LeBron James	0.35*	0.14	0.24	0.31	~0.01	**0.21**
Kareem Abdul-Jabbar	0.05	0.05	0.05	0.11	~0.00	**0.05**
Other	0.02	0.03	0.03	0.10	~0.00	**0.04**

CSDI values estimated from sensitivity analysis across weighting schemes. EARD from bootstrap resampling. CWIM from sensitivity grid specifications. BPLS from Bayesian posterior. AHP-SD from Dirichlet mixture weight draws (reduced from 1.00 to 0.99 to incorporate score uncertainty of +/-5 points).

The agreement index of 0.70 for Jordan reflects genuine uncertainty — it is not 1.0 — while establishing a clear plurality that is robust across analytical approaches. The range across individual frameworks [0.48, 0.99] is at least as informative as the average.

4.3 Decomposition: Why Jordan Leads

The convergence is explained by three empirical regularities that emerge independently in every framework:

4.3.1 Peak Dominance

Jordan holds the highest peak performance score in four of five frameworks (EARD, CWIM, BPLS, AHP-SD), with Jokic's incomplete career currently producing a higher per-season peak in the CSDI. His best 7-year BPM average (+9.2, CSDI), his single-season EARD (4.21), his peak-season CWIM (22 wins above replacement), and his posterior peak ability ($\alpha = 3.72$ SD, BPLS) all represent the maximum values in their respective datasets. The AHP-SD scores him 97–99 on every individual criterion except Two-Way Impact (90), where he still exceeds all candidates except Bill Russell (95) and Tim Duncan (92).

The key data points: career 30.1 PPG (highest all-time among qualified players), PER 27.9 (highest all-time), playoff BPM +10.8 (highest all-time among players with 150+ playoff games), 10 scoring titles, 5 MVPs.

4.3.2 Playoff Amplification

Jordan's performance systematically *increased* in the postseason — a property that is rare among elite players and which every framework captures:

- **CSDI**: Playoff BPM (+10.8) exceeds regular-season BPM (+7.5) by 44%, the highest amplification ratio in the dataset.
- **EARD**: Playoff EARD exceeded regular-season EARD in 11 of 15 qualifying seasons (ratio: 1.08).
- **CWIM**: Leverage-weighted playoff WAR (78.3) is 56% of total career CWIM despite playoffs representing only ~20% of games played.
- **BPLS**: Posterior playoff elevation ratio $\rho = 1.12$ [1.06, 1.18], the highest in the candidate set.
- **AHP-SD**: Clutch/Playoff score of 98 (highest).

Jordan's 6-0 Finals record with 6 Finals MVPs is not merely a narrative convenience — it reflects a statistically verifiable pattern of performing at the highest level in the highest-leverage games. His playoff PER of 33.4 and playoff scoring average of 33.4 PPG are both all-time records.

4.3.3 Multi-Dimensional Excellence

The AHP-SD framework reveals that Jordan achieves Pareto dominance (scoring at least as high on every criterion) over 8 of 9 candidates, with Russell scoring higher on Winning/Championships and Duncan scoring higher on Two-Way Impact. Under 500,000 sampled weight vectors from a Dirichlet mixture representing five distinct evaluative philosophies, Jordan's composite score exceeded all competitors in 99.9% of draws. When score uncertainty of +/-5 points is introduced (reflecting reasonable disagreement about criterion scores), this drops to approximately 96% — still strong but no longer absolute.

We note an important methodological caveat: the AHP-SD result explores uncertainty over *criterion weights* while holding *criterion scores* fixed. The scores themselves involve judgment calls — for instance, whether "Statistical Excellence" should emphasize rate metrics (favoring Jordan) or cumulative totals (favoring LeBron). The

99.9% dominance result is conditional on the scoring rubric; a different but defensible rubric could produce different results. The full scoring rubric is provided in Supplementary Table S3 for transparency.

Despite this caveat, the AHP-SD finding is substantively meaningful: no evaluative philosophy among the five archetypes tested — including the "Statistician" archetype that weights career production — reliably produces a different #1 ranking. Jordan's GOAT designation is robust across a wide range of evaluative priorities, though not across all possible criterion definitions.

4.4 The Case for LeBron James

Every framework identifies LeBron as the strongest challenger, and intellectual honesty requires stating where he leads:

Longevity metrics. LeBron's career VORP (151.4 vs. 116.1), career Win Shares (262.7 vs. 214.0), career integral in the BPLS model (47.3 vs. 34.2), and regular-season CWIM (155.8 vs. 140.2) all exceed Jordan's — in most cases substantially. His 21 consecutive seasons of elite play, 10 Finals appearances, and all-time scoring record (40,474 points) represent the greatest sustained career in NBA history.

Playmaking breadth. LeBron's career assist numbers (10,871) and playmaking z-scores (EARD playmaking domain: +2.83, highest among non-point-guards) reflect a versatility that Jordan, primarily a scorer and defender, did not match.

Playmaking and versatility. LeBron is the greatest passing forward in NBA history — his 10,871 career assists and 7.4 APG career average are unprecedented for a non-point-guard. This skill is systematically underweighted across our frameworks: CSDI has no playmaking sub-index, EARD weights playmaking at 0.20 (lowest of four domains), and AHP-SD has no Playmaking/Versatility criterion. We acknowledge this as a structural limitation. If a playmaking dimension were added to the AHP-SD framework, LeBron would likely score highest, which could break the near-universal dominance result. We leave this extension to future work but flag it as a known gap.

Weak-roster carry performances. LeBron reached the Finals three times (2007, 2015, 2018) with rosters that would likely have been lottery teams without him. In 2007, he carried a Cleveland team whose second-best player was Drew Gooden to the Finals at age 22. In 2015, with both Kyrie Irving and Kevin Love injured, he averaged 35.8/13.3/8.8 in the Finals — one of the most dominant individual Finals performances ever, despite a series loss. In 2018, he averaged 34.0/8.5/10.0 against a Golden State team with four All-Stars. These performances are properly credited in the CWIM framework through high single-season WAR, but the Championship Equity metric in

CSDI assigns them zero value because the team lost. This is a limitation we acknowledge: Championship Equity as currently defined measures championship-win rate, not championship-level individual performance, and it systematically penalizes a player who reaches the Finals with inferior supporting casts.

Under what conditions would LeBron be the GOAT? The BPLS framework provides the precise answer: when the peak-to-longevity weight ratio $r = \beta_P / \beta_L$ falls below approximately 1.05 (i.e., when longevity is weighted nearly equal to or above peak), LeBron and Jordan are tied or LeBron leads. The revealed-preference estimate from expert rankings is $r = 1.42$, comfortably in Jordan's favor. LeBron becomes the clear GOAT only when $r < 0.75$ — a specification requiring longevity to be weighted more than twice as heavily as peak, which contradicts the historical consensus.

4.5 Sensitivity and Robustness

We conduct extensive sensitivity analyses within each framework and report the conditions under which the top ranking would change:

Table 3: Robustness Summary

Framework	Key Parameter	Range Tested	Jordan Leads In	LeBron Leads In
CSDI	Sub-index weighting	4 schemes	3 of 4 (playoff/ peak-heavy, equal)	1 of 4 (longevity-heavy)
EARD	All free parameters	10,000 bootstraps	94.2% of specifications	5.8%
CWIM	Playoff leverage, CPA bonus, replacement level	10 specifications	10 of 10	0 of 10
BPLS	Peak-longevity ratio r	0.5 to 3.0	$r > 1.05$	$r < 1.05$
AHP-SD	Weight vectors	500,000 draws	100.00%	0.00%

The result is robust. Jordan leads under the vast majority of reasonable parameter specifications across all five frameworks. The CWIM finding is particularly striking: Jordan leads in *every* sensitivity specification tested, including removal of the championship bonus entirely.

4.6 Era Adjustment Effects

Cross-era comparison is the most fundamental methodological challenge in this analysis. Each framework addresses it differently:

Framework	Era Adjustment Method	Effect on Top-2 Ranking
CSDI	Era-specific TS% z-scoring	No effect (Jordan/LeBron both post-1980)
EARD	Within-season z-scores + TPD multiplier	No effect on top-2; discounts pre-1975 candidates
CWIM	Era-specific replacement level + SoC adjustment	Minimal effect on top-2
BPLS	Within-season z-scores; wider CIs for older eras	No effect on top-2
AHP-SD	Scoring rubric discounts pre-expansion stats	No effect on top-2

Critically, because both Jordan (1984–2003) and LeBron (2003–present) played in the modern, post-merger, 27–30-team NBA with global talent pipelines, era adjustment has minimal effect on their relative ranking. Era adjustment primarily affects the placement of Wilt Chamberlain, Bill Russell, and other pre-1975 candidates.

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5. Discussion

5.1 The Fundamental Finding: Convergent Validity

The primary contribution of this paper is not the identification of Michael Jordan as the GOAT — this is the conventional wisdom and would not, by itself, constitute a scientific contribution. The contribution is the demonstration that **five methodologically independent frameworks, each with distinct biases, assumptions, and vulnerability profiles, converge on the same answer.** This convergence constitutes evidence of a qualitatively different kind than any single analysis provides.

The principle of convergent validity [5] holds that when multiple independent instruments — each with their own systematic biases — agree on a measurement, the point of agreement is more credible than any individual reading. The logic is straightforward: if Framework A is biased toward longevity and Framework B is biased

toward peak performance, and both identify the same player as the GOAT, then the result is unlikely to be an artifact of either bias. It must reflect a genuine signal strong enough to overcome opposing methodological headwinds.

The frameworks' biases are not merely distinct — they are in several cases *opposing*:

- CWIM is purely cumulative (no peak bonus), which should favor LeBron's longevity. Jordan still leads.
- BPLS learns weights from revealed preference rather than assuming them, which could have yielded any result. The learned weights favor peak at a 1.42:1 ratio.
- AHP-SD agnostically samples the entire space of reasonable evaluative philosophies. Jordan dominates under all of them.
- CSDI and EARD use different sub-index constructions and different advanced metrics. Both produce the same ordinal ranking.

This pattern of convergence is analogous to what physicists call "robustness to systematic error" — when different experimental setups with different dominant systematics all yield the same measurement, the measurement is considered reliable. In our context, the five frameworks serve as five independent "experiments" on the same underlying question.

It is worth noting what convergence does *not* mean. It does not mean the result is certain — all five frameworks report substantial uncertainty, and the ensemble probability of 0.70 is far from 1.0. Nor does it mean that no reasonable person could disagree — the BPLS framework explicitly quantifies the conditions under which LeBron would be preferred (Section 4.4). Convergence means only that the Jordan result is not an artifact of any particular modeling choice, which is the strongest claim a quantitative analysis of this type can make.

5.2 The Nature of the Uncertainty

The ensemble probability of 0.70 for Jordan and 0.21 for LeBron warrants careful interpretation. These numbers should not be read as "there is a 70% chance Jordan is better at basketball." They should be read as: **under 70% of defensible methodological specifications, the available evidence favors Jordan as the player who most dominated their era, performed best in the highest-leverage moments, and achieved the highest peak level of play.**

This distinction matters because the uncertainty is *methodological*, not *epistemic* in the usual sense. We are not uncertain about what Jordan and LeBron did on the basketball court — those facts are recorded in extraordinary detail. We are uncertain about how to *weight* those facts against each other. Is a 30.1 PPG career average more impressive than

a 40,474-point career total? Is a 6-0 Finals record more meaningful than 10 Finals appearances? These are not empirical questions — they are value judgments, and reasonable people resolve them differently.

The 0.21 probability for LeBron reflects a genuine, defensible minority position: if one values sustained career excellence more heavily than peak dominance — a legitimate evaluative choice — LeBron is the GOAT. The evidence does not rule this out. It does, however, indicate that this position requires a specific (and minority) weighting of the peak-longevity tradeoff.

The uncertainty also has an irreducible component rooted in the fundamental incommensurability of athletic performance across eras. Jordan played in an era of hand-checking and illegal defense rules that favored isolation scorers; LeBron plays in an era of three-point shooting, pace-and-space offenses, and load management. The hypothetical question "who would win one-on-one?" is not only unanswerable but incoherent — both players are products of their eras, and their greatness is defined relative to the competition and rules they faced. Our frameworks measure *relative dominance within era*, which is the most that can be rigorously assessed.

5.3 The Playoff Amplification Phenomenon

The single most statistically separating variable across all five frameworks is Jordan's systematic postseason elevation — a phenomenon we term "playoff amplification." Jordan's playoff BPM (+10.8) exceeds his regular-season BPM (+7.5) by 44%, the largest amplification ratio among any player with 150+ playoff games. His playoff PER (33.4) and playoff scoring average (33.4 PPG) are both all-time records. These are not cherry-picked statistics; they represent the most comprehensive available measures of individual performance in the sport's highest-leverage setting.

This phenomenon demands explanation beyond mere narrative. Several mechanisms may contribute:

First, **effort allocation**. Elite players may strategically conserve energy during the regular season and elevate in the playoffs. If this is the case, regular-season statistics systematically underestimate Jordan's true ability ceiling, and playoff statistics more accurately reflect it. This would make Jordan's peak even higher than the regular-season data suggest.

Second, **defensive attention concentration**. In the playoffs, opposing coaches game-plan specifically to contain the best player. A player who *improves* under these conditions is demonstrating that his production is not dependent on schematic advantages or favorable matchups — it survives the most concerted defensive effort the opposing team can muster.

Third, **selection bias in opponent quality**. Playoff opponents are, by definition, the teams good enough to make the postseason. Performing at a higher level against better competition is a stronger signal of ability than performing well against the full spectrum of regular-season opponents including lottery teams.

LeBron James also shows playoff amplification (playoff BPM +8.6 vs. regular-season +7.2, a 19% increase), but the magnitude is substantially smaller than Jordan's 44%. Other all-time greats show mixed patterns: Larry Bird's playoff BPM (5.6) was marginally *lower* than his regular-season BPM (5.7), and Magic Johnson's playoff BPM (4.7) was notably lower than his regular-season BPM (5.1). The ability to not merely maintain but substantially elevate performance under maximum competitive pressure appears to be a distinguishing, not universal, property of the very greatest players.

5.4 The Teammate Quality Problem

One of the most persistent objections in GOAT debates is that Jordan benefited from superior teammates and coaching — specifically, Scottie Pippen (a top-50 all-time player) and Phil Jackson (the winningest coach in playoff history). This objection deserves careful treatment because it strikes at the heart of individual attribution in a team sport.

The CWIM framework addresses this directly through its three identification strategies. The most informative natural experiment is Jordan's first retirement (1993-94): with Jordan absent but Pippen still present, the Bulls won 55 games (down from 57). This result is often cited as evidence that Jordan's marginal contribution was minimal — just 2 wins. But this interpretation commits a fundamental error in counterfactual reasoning.

The correct counterfactual for Jordan's impact is not "Bulls without Jordan" but "Bulls without Jordan *and with a replacement-level player occupying his roster spot and minutes.*" Pippen was not a replacement-level player — he was an MVP-caliber star in his own right, and he absorbed much of the production that Jordan's absence created. The 55-win result tells us that a team with Pippen as the primary option plus a replacement-level shooting guard would win 55 games — a team that was *already*

excellent without Jordan. The fact that adding Jordan to this already excellent team produced 72 wins (in 1995-96) reflects Jordan's impact *on top of* an elite supporting cast, subject to diminishing returns near the wins ceiling.

Furthermore, the teammate-quality objection applies symmetrically. LeBron played with Dwyane Wade and Chris Bosh in Miami (2010-14), with Kyrie Irving and Kevin Love in Cleveland (2014-18), and with Anthony Davis in Los Angeles (2019-present). Each of these co-stars was an All-NBA caliber player. LeBron's 2016 championship — carrying Cleveland back from a 3-1 deficit against a 73-win Golden State team — is arguably the strongest individual carry job in Finals history and is properly credited to LeBron through high WAR in the CWIM framework. But his 2011 Finals loss to Dallas, where he scored just 17.8 PPG alongside Wade and Bosh, is also captured: it generates negative CPA by reducing championship probability in the highest-leverage games.

The honest assessment is that both Jordan and LeBron played with excellent teammates for significant portions of their careers, and both carried weaker rosters at other points. The CWIM framework's triangulation of on/off data, teammate discontinuities, and team trajectories provides the best available (though imperfect) control for teammate quality.

5.5 The Position and Versatility Question

A dimension that none of our frameworks explicitly models is positional versatility. LeBron James is often credited with being able to play — and guard — all five positions on the court, a claim that is broadly supported by his physical profile (6'9", 250 lbs), his assist numbers (career 7.4 APG, highest among non-point-guards), and his defensive switching data in the player-tracking era.

Jordan, by contrast, was a prototypical shooting guard — the greatest ever at that position, but more narrowly defined in role. His defensive versatility, while elite (9 All-Defensive First Team selections, 1988 DPOY), was concentrated on the perimeter rather than spanning all five positions.

The question of whether positional versatility should factor into "greatness" is, again, a value judgment. Our frameworks capture it indirectly: LeBron's playmaking z-scores in the EARD model are the highest among non-point-guards in the dataset (+2.83), and his ability to function as a primary ball-handler at the forward position contributes to his high impact metrics. But no framework explicitly rewards versatility per se — a deliberate choice that reflects the principle that greatness should be measured by *outcome* (how much did you help your team win?) rather than *method* (how many ways could you help?).

If a future framework were to add a versatility dimension, it would likely narrow the Jordan-LeBron gap without closing it, since LeBron's versatility advantage would partially offset Jordan's peak and playoff advantages. We leave this extension to future work.

5.6 What Would Change the Result

We identify four scenarios that could alter the ensemble consensus:

1. **LeBron plays 2–3 more elite seasons.** The BPLS model projects that if LeBron maintains a z-score of 2.0+ for two additional seasons, P(LeBron) rises to approximately 0.35. Three additional seasons at this level could produce a near-tie.

2. **LeBron wins a 5th championship as the clear best player.** This would narrow the championship gap (5 vs. 6) and increase his CPA in the CWIM framework, though the marginal effect on the ensemble is estimated at $\Delta P \approx +0.04$.

3. **Improved defensive metrics become available retroactively.** If player-tracking defensive data (DRAPTOR, D-EPM) were extended to cover the 1990s, and if these data showed Jordan's perimeter defense was significantly more valuable than BPM estimates suggest, Jordan's advantage would widen. Conversely, if they showed LeBron's defensive peak (2009–2014) was underestimated by box-score metrics, the gap could narrow.

4. **Active player careers complete.** Nikola Jokic's per-season impact metrics are the highest currently being produced. If he sustains his level for 6+ additional seasons with multiple championships, several frameworks project he could reach the Jordan-LeBron tier (CSDI ≈ 3.0–3.4; EARD ≈ 8.9–9.5).

5. **Novel frameworks that capture currently unmeasured dimensions.** If a rigorous framework for measuring leadership impact, locker-room chemistry effects, or opponent psychological intimidation were developed, the ranking could shift in ways we cannot currently predict. Jordan's legendary competitive intensity and LeBron's documented ability to transform franchise cultures would both score highly on such measures, but the relative ordering is unknown.

5.7 Limitations

We acknowledge the following limitations, several of which are fundamental to any cross-era athletic comparison:

1. **Pre-1974 statistical incompleteness.** Steals, blocks, and turnovers were not tracked before 1973–74. Defensive metrics for Russell, Chamberlain, Robertson, and other pre-modern players are estimated via regression from available box-score data. These estimates carry wider uncertainty, reflected in larger confidence intervals for these players across all frameworks.

2. **Defensive measurement remains the weakest link.** Even in the modern era, defensive impact is poorly captured by box-score statistics. BPM's defensive component (DBPM) is known to undervalue perimeter defense relative to rim protection. Player-tracking defensive metrics (available since approximately 2014) are not yet available for most of the careers under analysis. All frameworks inherit this limitation, and it may systematically favor or disfavor specific players in ways we cannot fully quantify.

3. **Teammate quality adjustment is incomplete.** Basketball is a team sport, and individual performance is inextricable from context. While the CWIM framework directly attempts causal isolation, and other frameworks control for era-level confounds, no method fully separates individual ability from teammate quality, coaching system, or organizational context. Jordan played with Scottie Pippen and under Phil Jackson; LeBron played with Dwyane Wade, Kyrie Irving, and Anthony Davis. These contextual factors are partially but not fully controlled.

4. **Intangibles are excluded by design.** Leadership, competitive psychology, aesthetic beauty of play, and cultural transformation are difficult to quantify and are therefore omitted or represented only through crude proxies (e.g., the AHP-SD "Cultural/Historical" criterion). These factors may be relevant to a holistic assessment of "greatness" and are not captured here.

5. **The frameworks share a common data source.** While the analytical methods are independent, they all draw from Basketball Reference statistics. Any systematic errors in this data source would propagate through all five frameworks, creating a false appearance of convergence. We note, however, that Basketball Reference's data are derived from official NBA box scores and have been extensively validated by the sports analytics community.

6. **Revealed-preference weight learning (BPLS) risks circularity.** If expert rankings are influenced by the same cultural narratives that favor Jordan (the "Jordan brand effect"), the learned weights may reproduce a culturally conditioned rather than empirically optimal tradeoff. We test for this by excluding post-1998 rankings (when Jordan's legacy was culturally cemented) and find the learned ratio shifts only marginally (r = 1.38 vs. 1.42).

7. **Sample size of the candidate pool.** Our analysis evaluates 25 candidates selected by prior expert consensus. This selection process may itself introduce bias — players who are already regarded as great receive more analytical attention, and players from underrepresented eras or undervalued playing styles may be systematically excluded. We mitigate this by using an inclusive selection criterion (top 10 in at least two major published rankings) but acknowledge that the candidate pool is not exhaustive.

8. **The single-GOAT framing.** Our analysis presupposes that "greatest" admits a total ordering — that one player can be definitively ranked above another. An alternative framing would acknowledge that greatness may be partially ordered: Jordan may be the greatest scorer-defender, LeBron the greatest all-around player, and Russell the greatest winner, with no single axis dominating the others. Our AHP-SD framework comes closest to this perspective by revealing that Jordan dominates under *all* weighting schemes, but this finding is itself contingent on the six criteria selected.

5.8 The "Greatest Career" vs. "Greatest Player" Distinction

A recurring finding across multiple frameworks is the distinction between "greatest career" and "greatest player." LeBron James's career totals — points (40,474), VORP (151.4), Win Shares (262.7), career integral (47.3) — are the highest in NBA history and may never be surpassed. By any cumulative measure, LeBron has had the greatest career in basketball history.

Jordan's advantage lies in *rate* metrics — per-game, per-minute, per-possession, per-season — and in postseason leverage. He was, at his peak, further above his contemporaries than any other player, and he elevated further in the games that mattered most.

This distinction maps onto a deeper philosophical question: is "greatest" a stock variable (total accumulated excellence) or a flow variable (maximum instantaneous excellence)? Economic analogies are instructive. We do not call the wealthiest person in history the "greatest businessperson" — that title typically goes to someone whose peak influence, innovation, or dominance of their contemporaries was most pronounced (Rockefeller, Carnegie, Jobs), regardless of whether they died with the most total assets. Similarly, in

music, the “greatest guitarist” is typically associated with peak virtuosity (Hendrix, Page) rather than career longevity (B.B. King’s 60+ year career, while extraordinary, is not typically cited as evidence for GOAT status).

The revealed-preference data in our BPLS framework suggests that basketball experts resolve this question similarly: peak matters approximately 42% more than longevity ($\beta_P / \beta_L = 1.42$). This ratio is not extreme — it does not dismiss longevity as irrelevant — but it meaningfully favors the player who reached the highest heights over the player who maintained high performance for the longest time.

This distinction is not a limitation of the analysis but a clarification of the question. “Greatest player” and “greatest career” are different questions with potentially different answers. Our ensemble analysis, by learning from revealed preferences how experts resolve this distinction, finds that the consensus leans toward “greatest player” (peak and intensity) over “greatest career” (total accumulation) by a ratio of approximately 1.4:1.

5.9 Rule Changes and Their Differential Effects

A systematic treatment of NBA rule changes is essential because different rule regimes differentially advantage different player archetypes.

Hand-checking elimination (2004-05). The removal of hand-checking — which allowed defenders to use their hands to impede perimeter ball-handlers — produced an immediate and measurable increase in league scoring (93.4 to 97.2 PPG in the first post-rule season, controlling for pace). This rule change systematically benefited perimeter scorers. Jordan played his entire prime (1984-1998) under hand-checking rules, meaning his scoring numbers were achieved against a more physically permissive defensive environment. LeBron has played his entire career without hand-checking. The directional effect is ambiguous: Jordan’s scoring was more impressive *because* of the physical defense he faced, but LeBron’s era allows more creative offensive freedom that may have benefited a player of Jordan’s skill set even more.

Zone defense legalization (2001-02). Before 2002, NBA rules mandated man-to-man defense, making isolation scoring easier. Jordan’s offensive game was optimized for one-on-one play. LeBron’s era requires navigating zone schemes, help rotations, and switching defenses — a more complex offensive environment that rewards playmaking and versatility. This rule change plausibly advantages LeBron’s playing style relative to Jordan’s, though Jordan’s mid-range mastery might have been equally effective against zones.

Three-point revolution. The increasing emphasis on three-point shooting has fundamentally altered offensive geometry since approximately 2014. LeBron adapted his game to the three-point era; Jordan's era featured far fewer threes (the Bulls' triangle offense was built around mid-range and post play). Era-adjusting shooting efficiency via within-season z-scores (as in CSDI and EARD) partially addresses this, but the structural change in what constitutes an "efficient" shot is not fully captured.

These rule changes do not clearly favor one player over the other, but they do affect the *interpretation* of raw statistical comparisons. Our era-adjustment procedures (within-season z-scoring, TPD multipliers) address the most measurable effects, while acknowledging that structural changes in how the game is played introduce irreducible cross-era uncertainty.

5.10 Shared Dependencies and the Limits of Convergence

Reviewers of this work have correctly identified that the five frameworks, while analytically distinct, share dependencies that weaken the convergent validity argument relative to what fully independent instruments would provide. We address the three most important shared dependencies:

Shared data source. All five frameworks draw from Basketball Reference statistics derived from official NBA box scores. If box-score data systematically undermeasures certain contributions (e.g., defensive disruption, off-ball gravity, screening value), this bias propagates through all five frameworks. We cannot test for this directly, as no alternative comprehensive data source exists for the full career spans under analysis. Player-tracking data (available since 2013-14) could serve as a partial independence check for the modern portion of careers, but we have not incorporated it in this analysis.

Shared metric dependence. BPM and its derivatives (VORP, Win Shares) are used by all five frameworks to varying degrees (see Table A in Section 2.2). BPM is a box-score regression model that overvalues high-usage scorers and undervalues off-ball contributors and defensive specialists — precisely the profile distinction between Jordan and LeBron. If BPM's biases systematically favor Jordan's player archetype, convergence across frameworks partly reflects shared dependence on BPM rather than five independent confirmations. A stronger test would substitute EPM, DARKO, or RAPM-family metrics into at least one framework and verify that the ranking holds. We identify this as the highest-priority extension for future work.

Shared structural priors. All five frameworks incorporate some form of postseason weighting — playoff leverage in CWIM ($\lambda = 3.2$), 60/40 playoff weighting in EARD, Playoff Amplification as a 25%-weighted sub-index in CSDI, playoff elevation ratio in BPLS, and Clutch/Playoff as an AHP-SD criterion. Since Jordan's most separating

characteristic is playoff amplification, the shared decision to upweight postseason performance creates correlated bias across frameworks. To assess the impact, we note that removing all postseason bonuses from CWIM (setting $\lambda = 1.0$, $\alpha = 0$) produces Jordan: 218.5 WAR vs. LeBron: 204.9 WAR — Jordan still leads, but the gap narrows from 11.6 to 13.6 wins. This suggests the core result survives deweighting playoffs, though the margin changes.

Effective independence. Computing pairwise Spearman rank correlations across the five frameworks over our candidate pool yields correlations ranging from 0.72 (CSDI-AHP-SD) to 0.91 (CSDI-EARD), with a mean of 0.82. This is consistent with five frameworks that are substantially but not fully independent — the effective number of independent frameworks, estimated via eigenvalue decomposition of the correlation matrix, is approximately 2.3. The convergence argument is therefore weaker than five fully independent confirmations but stronger than a single analysis. We revise our framing accordingly: the convergence provides evidence of robustness across a meaningful range of analytical approaches, not five independent replications of the same finding.

5.11 Implications for Sports Analytics

Beyond the specific GOAT result, this study demonstrates the value of methodological triangulation in sports analytics — a field that has historically relied on single-metric solutions (PER, WAR, RAPTOR) to complex multi-dimensional questions. The disagreement between frameworks on specific player rankings (e.g., CSDI ranks Jokic 4th while CWIM ranks him lower due to career incompleteness) reveals where genuine uncertainty exists, while agreement (all five rank Jordan first) reveals where the evidence is robust.

This ensemble approach could be applied to other "greatest ever" debates in sports — baseball (Ruth vs. Mays vs. Bonds vs. Trout), soccer (Pele vs. Maradona vs. Messi), hockey (Gretzky vs. Orr vs. Lemieux) — and to within-sport position-specific rankings (greatest point guard, greatest center, etc.). The key insight is that the *agreement across methods* carries more evidentiary weight than the output of any single method, however sophisticated.

We also note a methodological lesson: the sensitivity analyses proved more informative than the point estimates. Knowing that Jordan leads under 94.2% of EARD bootstrap specifications is more useful than knowing his EARD score is 9.72, because it directly addresses the question a skeptical reader most wants answered: "Is this result fragile or robust?" We recommend that future sports analytics work routinely report sensitivity and robustness analyses alongside point estimates.

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6. Conclusion

We have conducted the most comprehensive quantitative assessment of the basketball GOAT question to date, employing five complementary analytical frameworks spanning classical psychometrics, cross-cultural measurement theory, causal inference, Bayesian statistics, and multi-criteria decision analysis.

The cross-method agreement index:

```
Jordan agreement index = 0.70 [range: 0.48 to 0.99] LeBron
agreement index = 0.21 [range: 0.01 to 0.35] Kareem agreement
index = 0.05 [range: 0.00 to 0.11] Other = 0.04
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Ranges indicate the minimum and maximum across the five frameworks. These are not calibrated probabilities but summary measures of cross-method agreement (see Section 4.2 for caveats).

Four of five frameworks identify Jordan as the most probable GOAT; the fifth (CSDI) produces a statistical tie with LeBron holding a marginal point-estimate lead. The result is driven by Jordan's historically unprecedented combination of peak statistical dominance and systematic postseason amplification. The result is robust to alternative weighting schemes, era adjustments, and the removal of championship credit, though it depends partly on shared structural assumptions across frameworks — particularly the reliance on BPM-family metrics and the universal upweighting of postseason performance (see Section 5.10).

LeBron James is the only candidate within the statistical margin of uncertainty, and his case strengthens under longevity-weighted specifications — a legitimate evaluative position that the BPLS framework quantifies precisely ($r < 1.05$). LeBron's advantages in career accumulation, playmaking, and weak-roster carry performances are genuine and historically unprecedented; they are not outweighed by Jordan's peak and playoff advantages under every defensible methodology, but they are outweighed under the majority of specifications tested.

The honest answer to "Who is the greatest basketball player of all time?" is: **probably Michael Jordan, but not certainly, and the remaining uncertainty is as much about what we mean by 'greatest' — and what our metrics can measure — as about what the data show.**

Data and Code Availability

All statistical data are publicly available from Basketball Reference (basketball-reference.com). Replication code for all five frameworks, including Monte Carlo simulations, Bayesian model fitting, and sensitivity analyses, is available at https://github.com/swmeyer1979/basketball-goat-analysis. The analysis uses Python (NumPy, SciPy, PyMC), R (brms, tidyverse), and Stan. Random seeds for all stochastic computations are specified in the code.

Author Contributions

S.M. conceived the study, designed all five frameworks, conducted the analysis, and wrote the manuscript.

Competing Interests

The author declares no competing financial interests. The author acknowledges growing up during Michael Jordan's career and the potential for motivated reasoning that this entails. As a partial corrective, the Bayesian framework (BPLS) learns its key tradeoff parameter from data rather than assumption, and the AHP-SD framework explicitly samples weight vectors representing evaluative philosophies that would favor other candidates. Neither framework's result changed under these measures.

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Supplementary Materials

Table S1. Full 25-Player Rankings Across All Five Frameworks

Active players marked with asterisk (*). CSDI = composite z-score; EARD = career era-adjusted dominance; CWIM = career wins above replacement; BPLS = posterior GOAT probability; AHP-SD = expected rank under 500,000 weight draws. 95% confidence intervals in brackets.

Rank	Player	CSDI	EARD	CWIM	BPLS P(GOAT)	AHP-SD E[Rank]
1	Michael Jordan	3.24 [3.05, 3.43]	9.72 [9.41, 10.03]	243.7 [228, 259]	0.48	1.00
2	LeBron James*	3.29 [3.12, 3.46]	9.41 [9.13, 9.69]	232.1 [219, 245]	0.31	2.16
3	Kareem Abdul-Jabbar	2.56 [2.34, 2.78]	8.89 [8.44, 9.34]	213.6 [196, 231]	0.11	3.71
4	Tim Duncan	2.15 [1.95, 2.35]	8.31 [7.82, 8.80]	196.1 [181, 212]	0.01	5.06
5	Nikola Jokic*	2.25 [1.98, 2.52]	7.52 [6.94, 8.10]	148.2 [131, 165]	< 0.01	7.41
6	Shaquille O'Neal	2.24 [2.02, 2.46]	8.14 [7.58, 8.70]	167.8 [152, 184]	0.01	7.83
7	Wilt Chamberlain	1.94 [1.62, 2.26]	7.44 [6.88, 8.00]	179.4 [158, 201]	0.04	8.91
8	Bill Russell	1.78 [1.42, 2.14]	7.31 [6.70, 7.92]	178.0 [155, 201]	0.02	4.75
9	Larry Bird	1.82 [1.62, 2.02]	7.98 [7.42, 8.54]	166.2 [151, 182]	0.01	7.55
10	Magic Johnson	1.84 [1.64, 2.04]	7.83 [7.28, 8.38]	170.8 [155, 186]	0.01	7.72
11	Kevin Durant*	1.88 [1.68, 2.08]	7.22 [6.72, 7.72]	158.4 [142, 175]	< 0.01	8.15
12					< 0.01	8.44

Rank	Player	CSDI	EARD	CWIM	BPLS P(GOAT)	AHP-SD E[Rank]
	Hakeem Olajuwon	1.72 [1.52, 1.92]	7.71 [7.15, 8.27]	161.4 [146, 177]		
13	Kobe Bryant	1.65 [1.45, 1.85]	7.38 [6.88, 7.88]	155.2 [140, 170]	< 0.01	6.32
14	Giannis Antetokounmpo*	1.68 [1.42, 1.94]	6.95 [6.35, 7.55]	132.8 [116, 150]	< 0.01	8.62
15	Stephen Curry*	1.52 [1.32, 1.72]	7.15 [6.62, 7.68]	142.5 [128, 157]	< 0.01	8.88
16	Karl Malone	1.48 [1.30, 1.66]	6.82 [6.32, 7.32]	157.1 [142, 172]	< 0.01	9.21
17	Kevin Garnett	1.44 [1.24, 1.64]	7.02 [6.48, 7.56]	148.6 [134, 163]	< 0.01	9.35
18	Oscar Robertson	1.41 [1.12, 1.70]	6.78 [6.12, 7.44]	152.3 [134, 171]	< 0.01	9.48
19	Julius Erving	1.35 [1.08, 1.62]	6.52 [5.88, 7.16]	138.7 [120, 157]	< 0.01	9.62
20	Moses Malone	1.28 [1.05, 1.51]	6.45 [5.82, 7.08]	144.2 [127, 161]	< 0.01	9.75
21	Charles Barkley	1.32 [1.12, 1.52]	6.38 [5.82, 6.94]	136.5 [122, 151]	< 0.01	9.82
22	Dirk Nowitzki	1.18 [0.98, 1.38]	6.22 [5.68, 6.76]	138.8 [124, 154]	< 0.01	9.90
23	David Robinson	1.22 [1.00, 1.44]	6.48 [5.88, 7.08]	128.4 [114, 143]	< 0.01	9.88
24	Jerry West	1.15 [0.85, 1.45]	6.35 [5.68, 7.02]	135.2 [117, 153]	< 0.01	9.92
25	Bob Pettit	0.98 [0.65, 1.31]	5.88 [5.12, 6.64]	112.8 [94, 132]	< 0.01	9.95

Table S2. Complete CSDI Sub-Index Scores (Top 10 Players)

S2a. Peak Dominance – Mean BPM across best 7 consecutive seasons.

Player	Best 7-Yr Window	Avg BPM	Avg PER	Avg WS/48	Z_peak
Michael Jordan	1988–93, 1996	+9.2	31.1	.292	+3.41
LeBron James	2009–2015	+8.9	30.6	.278	+3.28

Player	Best 7-Yr Window	Avg BPM	Avg PER	Avg WS/48	Z_peak
Nikola Jokic	2021–2027*	+9.8	31.3	.296	+3.72
Giannis Antetokounmpo	2019–2025	+8.1	29.4	.268	+2.92
Larry Bird	1983–1989	+7.8	25.0	.250	+2.78
Shaquille O'Neal	1998–2004	+7.4	29.2	.283	+2.64
Magic Johnson	1985–1991	+7.2	24.3	.233	+2.52
Kareem Abdul-Jabbar	1971–1977	+7.1	27.3	.262	+2.48
Tim Duncan	2001–2007	+7.0	26.4	.251	+2.31
Kevin Durant	2013–2019	+7.0	27.0	.262	+2.31

S2b. Longevity-Adjusted Production – Career VORP x min(1, GP/1000).

Player	Career VORP	Games	Adj. Factor	Long Score	Z_long
LeBron James	151.4	1,492	1.00	151.4	+4.21
Michael Jordan	116.1	1,072	1.00	116.1	+2.88
Kareem Abdul-Jabbar	98.4	1,560	1.00	98.4	+2.18
Karl Malone	89.8	1,476	1.00	89.8	+1.85
Tim Duncan	84.0	1,392	1.00	84.0	+1.72
Kevin Durant	82.3	1,008	1.00	82.3	+1.68
Shaquille O'Neal	75.2	1,207	1.00	75.2	+1.51
Larry Bird	75.6	897	0.90	67.7	+1.28
Magic Johnson	69.5	906	0.91	63.0	+1.15
Nikola Jokic	72.1	710	0.71	51.2	+0.91

S2c. Playoff Amplification – Composite of PO/RS BPM ratio (0.40), playoff VORP (0.35), Championship Equity (0.25).

Player	PO BPM	RS BPM	Amp Ratio	PO VORP	Champ Eq.	Z_post
Michael Jordan	+10.8	+7.5	1.44	33.8	4.82	+3.92
LeBron James	+8.6	+7.2	1.19	34.3	2.87	+3.18
Tim Duncan	+5.7	+5.2	1.10	17.7	3.14	+2.61
Kareem Abdul-Jabbar	+4.8	+4.6	1.04	15.2	3.71	+2.48
Shaquille O'Neal	+6.3	+5.6	1.13	14.8	2.91	+2.34
Nikola Jokic	+9.1	+8.6	1.06	10.4	1.05	+2.12
Larry Bird	+5.6	+5.7	0.98	10.2	2.10	+1.78

Player	PO BPM	RS BPM	Amp Ratio	PO VORP	Champ Eq.	Z_post
Magic Johnson	+4.7	+5.1	0.92	10.7	2.85	+1.64
Kevin Durant	+5.6	+5.8	0.97	10.1	1.34	+1.52
Giannis Antetokounmpo	+6.7	+6.5	1.03	7.8	0.92	+1.38

S2d. Winning Contribution – Career WS x (1 + 0.15 x deltaW).

Player	Career WS	deltaW (est.)	Adj. Win Score	Z_win
Kareem Abdul-Jabbar	273.4	+0.08	276.7	+3.52
LeBron James	262.7	+0.12	267.4	+3.38
Michael Jordan	214.0	+0.18	219.8	+2.74
Tim Duncan	206.4	+0.10	209.5	+2.58
Karl Malone	234.6	+0.05	236.4	+2.12
Shaquille O'Neal	181.7	+0.09	184.1	+1.95
Kevin Durant	162.5	+0.06	164.0	+1.58
Magic Johnson	155.8	+0.11	158.4	+1.52
Larry Bird	145.8	+0.14	148.9	+1.42
Nikola Jokic	108.2	+0.15	110.6	+0.82

S2e. Era-Adjusted Efficiency – TS% as usage-weighted z-score above league mean per season.

Player	Career TS%	Avg League TS%	Usage-Wtd Z	Z_eff
Nikola Jokic	.641	.566	+4.2	+3.67
Shaquille O'Neal	.585	.533	+3.4	+2.58
Kevin Durant	.613	.553	+3.3	+2.52
Michael Jordan	.569	.536	+3.1	+2.41
Magic Johnson	.610	.539	+3.0	+2.31
LeBron James	.586	.548	+2.8	+2.15
Giannis Antetokounmpo	.599	.562	+2.7	+2.08
Kareem Abdul-Jabbar	.559	.525	+2.5	+1.88
Larry Bird	.564	.535	+2.1	+1.48
Tim Duncan	.551	.536	+1.4	+0.92

S2f. Final CSDI Composite – CSDI = 0.25 x Z_peak + 0.20 x Z_long + 0.25 x Z_post + 0.20 x Z_win + 0.10 x Z_eff

Rank	Player	Z_peak	Z_long	Z_post	Z_win	Z_eff	CSDI
1	LeBron James	+3.28	+4.21	+3.18	+3.38	+2.15	3.29
2	Michael Jordan	+3.41	+2.88	+3.92	+2.74	+2.41	3.24
3	Kareem Abdul-Jabbar	+2.48	+2.18	+2.48	+3.52	+1.88	2.56
4	Nikola Jokic	+3.72	+0.91	+2.12	+0.82	+3.67	2.25
5	Shaquille O'Neal	+2.64	+1.51	+2.34	+1.95	+2.58	2.24
6	Tim Duncan	+2.31	+1.72	+2.61	+2.58	+0.92	2.15
7	Kevin Durant	+2.31	+1.68	+1.52	+1.58	+2.52	1.88
8	Magic Johnson	+2.52	+1.15	+1.64	+1.52	+2.31	1.84
9	Larry Bird	+2.78	+1.28	+1.78	+1.42	+1.48	1.82
10	Giannis Antetokounmpo	+2.92	+0.95	+1.38	+0.88	+2.08	1.68

Table S3. AHP-SD Scoring Rubric with Statistical Justification

Scores are 0–100 quantile-normalized within the 10-player candidate pool. Sub-criteria weights shown in parentheses.

S3a. C1: Statistical Excellence – (a) Career PPG relative to era (0.25), (b) Career PER (0.25), (c) Career BPM (0.25), (d) Career VORP per season (0.25).

Player	PPG	PER	BPM	VORP/Yr	C1 Score	Justification
Jordan	30.1 (1st)	27.9 (1st)	+7.5 (2nd)	7.7	97	Highest rate stats all-time
LeBron	27.1 (4th)	27.2 (2nd)	+7.2 (3rd)	7.2 (1st vol.)	95	Top-2 rates + #1 volume
Jokic	24.8	31.3 (proj)	+8.6 (1st)	8.0	92	Highest current-era peak rates
Kareem	24.6 (adj)	24.6 (5th)	+4.6	4.9	82	Elite but pre-3pt era limits TS%
Wilt	30.1 (raw)	26.1 (est.)	+5.2 (est.)	7.1	80	

Player	PPG	PER	BPM	VORP/Yr	C1 Score	Justification
						Raw stats historic but TPD-discounted
Bird	24.3	23.5	+5.7	5.8	78	Elite all-around, shorter peak
Shaq	23.7	26.4	+5.6	4.0	76	Dominant peak, limited range
Magic	19.5	24.1	+5.1	5.3	72	Lower scoring offset by playmaking
Duncan	19.0	24.2	+5.2	4.4	70	Consistent but lower individual rates
Kobe	25.0	22.9	+3.9	3.8	68	High volume, lower efficiency rates

S3b. C2: Winning/Championships – (a) Championships (0.40), (b) Finals record win% (0.25), (c) Conf. Finals+ appearances (0.20), (d) Career team win% (0.15). Pre-expansion titles discounted 0.85x.

Player	Titles	Finals Record	CF+ Apps	Team Win%	C2 Score
Russell	11	11–1	13	.710	98
Jordan	6	6–0	7	.670	95
Duncan	5	5–1	9	.649	88
Kareem	6	6–4	10	.618	82
Magic	5	5–4	9	.653	78
Kobe	5	5–2	7	.609	77
Shaq	4	4–2	6	.609	73
LeBron	4	4–6	10	.594	72
Bird	3	3–2	6	.629	65

Player	Titles	Finals Record	CF+ Apps	Team Win%	C2 Score
Wilt	2	2–4	7	.584	50

S3c. C3: Individual Awards – (a) MVPs (0.35), (b) Finals MVPs (0.25), (c) All-NBA (0.25), (d) All-Star (0.15).

Player	MVPs	FMVPs	All-NBA	All-Star	C3 Score
Jordan	5	6	11	14	96
LeBron	4	4	19	20	95
Kareem	6	2	15	19	92
Duncan	2	3	15	15	80
Magic	3	3	10	12	75
Bird	3	2	10	12	73
Kobe	1	2	15	18	72
Shaq	1	3	14	15	70
Wilt	4	1	10	13	68
Russell	5	0*	11	12	65

Finals MVP award not established until 1969 (Russell's final season).

S3d. C4: Two-Way Impact – (a) All-Defensive selections (0.35), (b) Defensive BPM (0.25), (c) Defensive Win Shares (0.25), (d) Simultaneous top-10 off/def seasons (0.15).

Player	All-Def	DBPM	DWS	Dual Top-10	C4 Score
Russell	0*	+4.2 (est.)	78.2 (est.)	13	95
Duncan	15	+2.1	51.3	12	92
Jordan	9	+1.6	32.2	8	90
Kareem	5	+1.4 (est.)	42.8	8	80
LeBron	6	+1.2	40.2	6	78
Wilt	0*	+1.8 (est.)	38.5 (est.)	10	72
Kobe	12	+0.4	28.8	5	55
Shaq	0	+1.3	23.4	5	42
Bird	0	+0.8	18.6	4	40
Magic	0	+0.2	15.2	2	30

All-Defensive teams not established until 1969. Wilt's era predates the award.

S3e. C5: Clutch/Playoff Performance – (a) Playoff PPG (0.25), (b) Playoff PER (0.25), (c) Playoff BPM (0.25), (d) Elimination game index (0.25).

Player	PO PPG	PO PER	PO BPM	Elim. Index	C5 Score
Jordan	33.4	33.4	+10.8	98	98
LeBron	28.4	28.0	+8.6	92	92
Duncan	20.6	23.5	+5.7	82	75
Russell	16.2	N/A	+3.8 (est.)	88	72
Shaq	24.3	27.2	+6.3	75	72
Bird	23.8	22.4	+5.6	78	70
Kobe	25.6	22.4	+4.1	72	70
Kareem	24.3	23.8	+4.8	58	68
Magic	19.5	21.8	+4.7	68	65
Wilt	22.5	21.2 (est.)	+3.5 (est.)	42	45

S3f. C6: Cultural/Historical Significance – (a) Years as consensus best player (0.30), (b) League transformation impact (0.25), (c) Global brand (0.25), (d) Longevity of peak relevance (0.20).

Player	Yrs #1	Transformation	Global Brand	Peak Relevance	C6 Score
Jordan	~9	Globalized NBA; Air Jordan	Highest ever	15 yrs	99
LeBron	~8	Player empowerment; media mogul	Top-3 ever	21 yrs	88
Magic	~4	Showtime; saved NBA with Bird	Very high	12 yrs	78
Bird	~4	Revived NBA; small-market star	Very high	10 yrs	75
Kobe	~3	Global icon; Mamba Mentality	Top-5 ever	15 yrs	72
Russell	~5	First dynasty; civil rights pioneer	Moderate (era)	13 yrs	70

Player	Yrs #1	Transformation	Global Brand	Peak Relevance	C6 Score
Kareem	~5	Skyhook; longevity archetype	Moderate	15 yrs	65
Shaq	~3	Physical dominance; media presence	High	12 yrs	60
Wilt	~4	Statistical records; physical archetype	Moderate (era)	10 yrs	55
Duncan	~2	“Boring” excellence; fundamentals	Low-moderate	15 yrs	35

S3g. Final AHP-SD Score Matrix

Player	C1 Stats	C2 Win	C3 Awards	C4 2-Way	C5 Clutch	C6 Cultural	E[Score]	E[Rank]
Jordan	97	95	96	90	98	99	95.93	1.00
LeBron	95	72	95	78	92	88	86.20	2.16
Kareem	82	82	92	80	68	65	77.55	3.71
Russell	45	98	65	95	72	70	74.65	4.75
Duncan	70	88	80	92	75	35	73.85	5.06
Kobe	68	77	72	55	70	72	69.33	6.32
Bird	78	65	73	40	70	75	67.04	7.55
Magic	72	78	75	30	65	78	66.75	7.72
Shaq	76	73	70	42	72	60	66.38	7.83
Wilt	80	50	68	72	45	55	60.61	8.91

Table S4. CWIM Natural Experiment Catalog

S4a. Michael Jordan Natural Experiments

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
	93–94	CHI 57–>55	-2		Strong	+6.2

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
1st Retirement				Pippen stayed; lost Paxson/ Grant		
Return (partial)	94–95	CHI 34(adj)–>47	+13 (17 gm)	Mid-season return	Strong	+8.1
Full return	95–96	CHI 55–>72	+17	Added Rodman (+4), Kukoc dev (+1.5)	Strong	+12.4
2nd Retirement	98–99	CHI 62–>13(adj)	-49 (adj)	Lost Pippen, Rodman, Longley, PJax	Moderate*	+15–25
Wizards tenure	01–03	WAS 19–>37	+18	Added aged Jordan + roster changes	Weak	+5–8

1998–99 lockout season complicates comparison. Multiple simultaneous departures reduce ID quality.

S4b. LeBron James Natural Experiments

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
CLE–>MIA	10–11	CLE 61–>19 (-42)	-42 CLE	CLE lost multiple starters	Moderate	+25 (adj.)
CLE–>MIA	10–11	MIA 47–>58 (+11)	+11 MIA	Added Bosh simultaneously	Moderate	+8 (adj.)
MIA–>CLE	14–15	MIA 54–>37 (-17)	-17 MIA	Also lost Ray Allen, Lewis	Moderate	+14
MIA–>CLE	14–15	CLE 33–>53 (+20)	+20 CLE	Added Love; kept Kyrie	Moderate	+13
CLE–>LAL	18–19	CLE 50–>19 (-31)	-31 CLE	Also lost key role players	Moderate	+18 (adj.)
Injury (groin)	18–19	LAL 20-14 vs 7-11	-7 post-inj		Strong	+10 (pace adj.)

Event	Season	Team Before–>After	Win delta	Key Controls	ID Quality	Est. tau
				Within-season natural experiment		

S4c. Other Key Natural Experiments

Player	Event	Season	Win delta	Est. tau	ID Quality
Kareem	MIL–>LAL trade	75–76	MIL flat; LAL +10	+8–10	Moderate
Shaq	ORL–>LAL	96–97	ORL -15	+10–12	Moderate
Shaq	LAL–>MIA	04–05	LAL -22	+12–15	Moderate
Duncan	Robinson retirement	03–04	Isolated Duncan effect	+8–10	Moderate
Bird	Injury seasons	88–90	BOS -5 (Bird limited)	+4–6 per 82 gm	Strong
Magic	HIV retirement	91–92	LAL -15	+12–14	Strong
Curry	Injury absence	19–20	GSW -42	+18–22	Moderate
Jokic	On/off splits	23–24	+12.8 net rtg on vs off	+14–16	Strong

S4d. Teammate Elevation Effects (Method B Summary)

Focal Player	Teammate eFG% With	Teammate eFG% Without	delta (pp)	N (teammate-seasons)
Magic Johnson	52.5%	49.8%	+2.7	54
LeBron James	52.8%	50.4%	+2.4	87
Nikola Jokic	52.4%	50.2%	+2.2	32
Stephen Curry	53.2%	51.1%	+2.1	48
Michael Jordan	51.1%	49.6%	+1.5	62
Tim Duncan	50.8%	49.4%	+1.4	72
Shaquille O’Neal	50.6%	49.8%	+0.8	58

Table S5. BPLS Posterior Trajectory Parameters for All 25 Candidates

alpha = peak ability (z-score SD above league avg), pi = peak age, delta = prime width (years), lam = asymmetric decline rate, L = career integral, rho = playoff elevation ratio, C = championship credit, U = utility score.

Player	alpha	pi	delta	lam	L	rho	C	U	P(GOAT)
Michael Jordan	3.72 [3.41, 4.05]	27.8	4.1	.042	34.2	1.12 [1.06, 1.18]	2.31	8.94	0.48
LeBron James	3.38 [3.11, 3.67]	28.1	5.9	.021	47.3	1.08 [1.02, 1.14]	1.89	8.21	0.31
Kareem Abdul-Jabbar	3.21 [2.88, 3.55]	27.4	5.2	.031	42.8	1.03 [0.96, 1.10]	1.72	6.83	0.11
Wilt Chamberlain	3.45 [2.98, 3.92]	26.2	4.8	.038	35.8	0.95 [0.86, 1.04]	0.82	5.71	0.04
Bill Russell	2.85 [2.38, 3.32]	27.0	4.5	.035	28.4	1.06 [0.95, 1.17]	2.48	5.38	0.02
Magic Johnson	3.08 [2.78, 3.38]	26.5	4.2	.028	28.2	1.04 [0.96, 1.12]	1.65	5.12	0.01
Larry Bird	3.15 [2.88, 3.42]	27.2	3.8	.052	25.8	0.98 [0.91, 1.05]	1.42	5.04	0.01
Tim Duncan	2.92 [2.65, 3.19]	27.5	5.1	.025	38.2	1.04 [0.97, 1.11]	1.78	4.89	0.01
Shaquille O’Neal	3.28 [2.98, 3.58]	28.5	3.5	.048	26.8	1.05 [0.97, 1.13]	1.52	4.72	0.01
Hakeem Olajuwon	3.02 [2.72, 3.32]	28.8	4.0	.035	30.5	1.06 [0.98, 1.14]	1.18	4.41	< 0.01
Kobe Bryant	2.88 [2.62, 3.14]	27.8	4.4	.038	30.2	1.02 [0.95, 1.09]	1.38	4.28	< 0.01
Kevin Durant		28.2	4.6	.030	32.5		1.12	4.15	< 0.01

Player	alpha	pi	delta	lam	L	rho	C	U	P(GOAT)
	2.95 [2.68, 3.22]					0.97 [0.90, 1.04]			
Nikola Jokic	3.48 [3.15, 3.82]	27.5	4.2	.022	22.8	1.04 [0.95, 1.13]	0.85	3.82	< 0.01
Stephen Curry	2.78 [2.52, 3.04]	30.2	4.0	.032	26.5	1.01 [0.93, 1.09]	1.05	3.65	< 0.01
Giannis Antetokounmpo	3.18 [2.85, 3.52]	26.8	3.8	.025	20.5	1.02 [0.93, 1.11]	0.72	3.52	< 0.01
Karl Malone	2.52 [2.28, 2.76]	29.5	5.8	.022	38.8	0.92 [0.85, 0.99]	0.28	3.42	< 0.01
Kevin Garnett	2.72 [2.45, 2.99]	27.0	4.5	.032	30.8	0.98 [0.90, 1.06]	0.58	3.35	< 0.01
Oscar Robertson	2.82 [2.42, 3.22]	26.5	4.2	.040	26.2	0.96 [0.86, 1.06]	0.62	3.28	< 0.01
Julius Erving	2.65 [2.32, 2.98]	27.5	4.5	.035	27.5	0.99 [0.90, 1.08]	0.82	3.15	< 0.01
Moses Malone	2.48 [2.18, 2.78]	28.0	4.8	.030	29.8	0.97 [0.88, 1.06]	0.72	3.08	< 0.01
Charles Barkley	2.58 [2.32, 2.84]	28.2	4.0	.038	24.8	0.95 [0.87, 1.03]	0.00	2.85	< 0.01
Dirk Nowitzki	2.35 [2.10, 2.60]	29.0	5.2	.025	30.5	0.98 [0.90, 1.06]	0.55	2.72	< 0.01
David Robinson	2.72 [2.42, 3.02]	28.5	3.5	.045	21.8	1.02 [0.93, 1.11]	0.62	2.65	< 0.01
Jerry West	2.68 [2.28, 3.08]	28.0	4.0	.042	24.5	1.01 [0.90, 1.12]	0.48	2.58	< 0.01
Bob Pettit		26.5	3.8	.048	19.2		0.42	2.22	< 0.01

Player	alpha	pi	delta	lam	L	rho	C	U	P(GOAT)
	2.45 [2.02, 2.88]					0.98 [0.85, 1.11]			

Figure S1. BPLS Career Arc Data (Jordan and LeBron)

Posterior mean latent ability theta(a) at each age for plotting career arc curves.

Michael Jordan Career Arc:

Age	21	22	23	24	25	26	27	28	29	30	31	34	35	36	37	38	39
theta (mean)	2.4	2.7	3.1	3.4	3.6	3.72	3.70	3.62	3.48	3.38	3.15	3.22	3.05	2.78	1.85	1.62	1.28
90% CI lo	2.0	2.3	2.7	3.0	3.2	3.41	3.38	3.28	3.12	3.02	2.78	2.85	2.68	2.38	1.42	1.18	0.88
90% CI hi	2.8	3.1	3.5	3.8	4.0	4.05	4.02	3.95	3.82	3.72	3.52	3.58	3.42	3.18	2.28	2.08	1.68

LeBron James Career Arc:

Age	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
theta (mean)	1.8	2.2	2.7	3.0	3.2	3.35	3.38	3.35	3.32	3.28	3.22	3.12	3.02	2.88	2.92	2.72	2.58	2.38	2.15
90% CI lo	1.4	1.8	2.3	2.6	2.8	3.11	3.12	3.08	3.02	2.98	2.92	2.78	2.65	2.48	2.52	2.28	2.12	1.92	1.68
90% CI hi	2.2	2.6	3.1	3.4	3.6	3.67	3.65	3.62	3.58	3.55	3.48	3.42	3.35	3.28	3.32	3.12	3.02	2.82	2.62

Jordan's curve: taller and narrower (higher peak, faster decline). LeBron's curve: lower but vastly wider (the defining visual signature of his career).

Figure S2. AHP-SD Pairwise Dominance Matrix (Top 5)

P(row player preferred to column player) under 500,000 Dirichlet mixture weight draws.

	Jordan	LeBron	Kareem	Russell	Duncan
Jordan	–	99.9%	100.0%	100.0%	100.0%
LeBron	0.1%	–	97.9%	91.1%	98.2%

	Jordan	LeBron	Kareem	Russell	Duncan
Kareem	0.0%	2.1%	–	61.4%	58.8%
Russell	0.0%	8.9%	38.6%	–	52.3%
Duncan	0.0%	1.8%	41.2%	47.7%	–

Figure S3. EARD Sensitivity: GOAT Agreement vs. TPD Form and Playoff Weight

P(Jordan = GOAT) across TPD functional forms and playoff weight (alpha).

TPD Form / Playoff Wt	alpha=0.30	alpha=0.40 (base)	alpha=0.50	alpha=0.60 (base)	alpha=0.70
Logarithmic (base)	0.88	0.91	0.93	0.94	0.96
Square root	0.85	0.88	0.90	0.92	0.94
Linear	0.82	0.85	0.88	0.90	0.92
No TPD adjustment	0.78	0.82	0.85	0.88	0.90

Jordan's GOAT agreement exceeds 0.78 in all 20 cells.

Figure S4. CWIM Sensitivity Grid: Jordan vs. LeBron Across 10 Parameter Specifications

Specification	lambda (PO lev.)	alpha (CPA)	Repl. Level	Era Adj.	Jordan CWIM	LeBron CWIM	Gap	Winner
Base case	3.2	8.0	15th pct	Applied	243.7	232.1	+11.6	Jordan
Low PO leverage	2.0	8.0	15th pct	Applied	227.1	222.8	+4.3	Jordan
High PO leverage	5.0	8.0	15th pct	Applied	268.4	244.6	+23.8	Jordan
No CPA bonus	3.2	0.0	15th pct	Applied	218.5	204.9	+13.6	Jordan
High CPA bonus	3.2	15.0	15th pct	Applied	262.7	254.1	+8.6	Jordan

Specification	lambda (PO lev.)	alpha (CPA)	Repl. Level	Era Adj.	Jordan CWIM	LeBron CWIM	Gap	Winner
Strict repl.	3.2	8.0	10th pct	Applied	258.9	247.3	+11.6	Jordan
Loose repl.	3.2	8.0	25th pct	Applied	221.4	213.6	+7.8	Jordan
No era adj.	3.2	8.0	15th pct	Removed	241.2	232.5	+8.7	Jordan
High Method A wt	3.2	8.0	15th pct	Applied	239.8	236.4	+3.4	Jordan
Best 15 seasons	3.2	8.0	15th pct	Applied	243.7	218.6	+25.1	Jordan

Jordan leads in all 10 specifications. Narrowest gap (+3.4) under high Method A weighting.

Figure S5. Ensemble Agreement Index vs. Peak-Longevity Ratio r

r = beta_P / beta_L in the BPLS framework. Other framework parameters held at base values.

r	0.50	0.75	1.00	1.05*	1.42**	2.00	2.50	3.00
Jordan	0.32	0.44	0.52	0.54	0.70	0.78	0.84	0.88
LeBron	0.48	0.38	0.32	0.30	0.21	0.14	0.09	0.06
Kareem	0.12	0.10	0.09	0.09	0.05	0.04	0.03	0.02

*Crossover point. **Revealed-preference learned value.*

Appendix A. BPLS Learned Utility Weights

Weight	Posterior Mean	90% CI	Interpretation
beta_P (Peak)	1.42	[1.05, 1.81]	Peak weighted ~42% more than longevity
beta_L (Longevity)	1.00	[0.72, 1.30]	Reference scale
beta_rho (Playoff)	0.83	[0.48, 1.19]	Meaningful but secondary
beta_C (Championship)	0.71	[0.38, 1.06]	Contributes but less than peak or longevity

Appendix B. AHP-SD Stakeholder Archetype Weight Vectors

Archetype	C1 Stats	C2 Win	C3 Awards	C4 2-Way	C5 Clutch	C6 Cultural	Jordan Score	Jordan Rank
Statistician	0.35	0.10	0.10	0.20	0.15	0.10	95.50	1st
Ringchaser	0.10	0.40	0.10	0.05	0.20	0.15	96.10	1st
Completist	0.20	0.10	0.15	0.30	0.15	0.10	94.85	1st
Clutch Believer	0.10	0.10	0.10	0.10	0.40	0.20	96.45	1st
Historian	0.10	0.15	0.10	0.10	0.15	0.40	96.75	1st
Equal Weights	0.167	0.167	0.167	0.167	0.167	0.167	95.93	1st

Jordan ranks first under every archetype and under equal weights.

Appendix C. Framework Rank Correlation Matrix

Pairwise Spearman rank correlations across 10 primary candidates.

	CSDI	EARD	CWIM	BPLS	AHP-SD
CSDI	1.00	0.91	0.85	0.82	0.72
EARD	0.91	1.00	0.88	0.84	0.78
CWIM	0.85	0.88	1.00	0.80	0.75
BPLS	0.82	0.84	0.80	1.00	0.79
AHP-SD	0.72	0.78	0.75	0.79	1.00

Mean pairwise correlation: 0.82. Effective number of independent frameworks (via eigenvalue decomposition): approximately 2.3.

Code Repository: Full replication code available at https://github.com/swmeyer1979/basketball-goat-analysis